Universität Zürich[UZH]
Center for Reproducible Science and Research Synthesis

Lotteries and innovations in research funding allocation

SORTEE webinar – July 07, 2026

Rachel Heyard

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Slides: https://osf.io/ujdah

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- **Disclaimer**: Open Science enthusiast, actively involved in the Swiss Reproducibility Network and CoARA (Coalition for Advancing Research Assessment)

My work in the SNSF data team

My work in the SNSF data team, illustrated via Gender Stats

Communicating the effect of gender in research funding

My work in the SNSF data team, illustrated via Gender Stats

Communicating the effect of gender in research funding

Three interacting bodies:

- Gender Equality Commission: independent advisory body with international members
- Gender Equality Office: supports internal bodies in implementing necessary measures
- Data Team: statisticians and data scientists supporting the office

My work in the SNSF data team, illustrated via Gender Stats

Communicating the effect of gender in research funding

Three interacting bodies:

- Gender Equality Commission: independent advisory body with international members
  - *advises* Gender Equality Office
- Gender Equality Office: supports internal bodies in implementing necessary measures
  - *reports to* Gender Equality Commission and mandates analyses to Data Team
- Data Team: statisticians and data scientists supporting the office

Lockdown: Are women submitting fewer grant proposals?

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From COVID "gender publication gap" to "gender application gap"

NEWS | 20 May 2020

Are women publishing less during the pandemic? Here's what the data say

Early analyses suggest that female academics are posting fewer preprints and starting fewer research projects than their male peers.

Gender Disparity in the Authorship of Biomedical Research Publications During the COVID-19 Pandemic: Retrospective Observational Study

Goran Muric [1]; Kristina Lerman [1, 2]; Emilio Ferrara [1, 2, 3]

Longitudinal analyses of gender differences in first authorship publications related to COVID-19

Carolin Lerchenmüller [1, 2], Leo Schmallenbach [3], Anupam B Jena [4, 5, 6], Marc J Lerchenmueller [3]

Gender inequality in publishing during the COVID-19 pandemic

Alana K. Ribarovska,[a] Mark R. Hutchinson,[b,c] Quentin J. Pittman,[d] Carmine Pariante,[e] and Sarah J. Spencer[a,b,*]

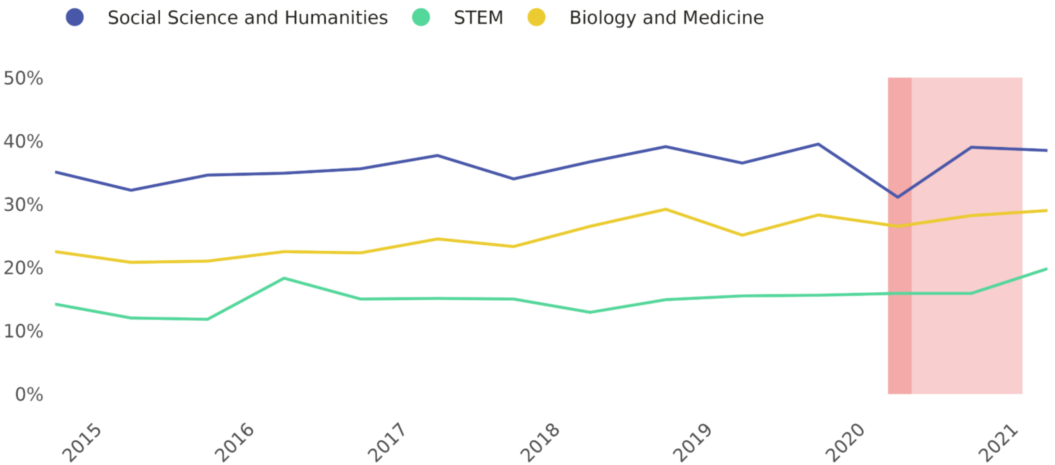
Lockdown: Are women submitting fewer grant proposals?

Analysed the share of female applicants applying to funding schemes with deadlines during or shortly after the lockdown.

Note:

- Lockdown with school closures in March 2020
- Postponement of Project Funding deadline from the 1 to 8 April

Female shares: Normal Project Funding



https://data.snf.ch/stories/women-submitting-fewer-grant-proposals-en.html

What is evidence-based science policy making?

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Evidence-based policy

Evidence-based policy refers to the practice of informing public policy decisions through the use of scientific evidence and rigorous research, a concept that has its roots in evidence-based medicine.
Source: https://www.ebsco.com/research-starters/social-sciences-and-humanities/evidence-based-policy

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Science policy

Science policy is concerned with the allocation of resources for the conduct of science towards the goal of best serving the public interest. Topics include the funding of science, the careers of scientists, and the translation of scientific discoveries into technological innovation to promote commercial product development, competitiveness, economic growth and economic development.
Source: Wikipedia

Research funding decisions

Focus on **competitive funding** by national and international scientific funding organisations

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$\Rightarrow$ They shape the research landscape: which research is done, and by whom?

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Focus on **competitive funding** by national and international scientific funding organisations

$\Rightarrow$ They shape the research landscape: which research is done, and by whom?

$\Rightarrow$ Lack of research & innovation going into decision-making **processes** (lack of evidence based policy making)

A standard funding decision-making process

Stage 0 - Funder designs the call	***Stage 1 -*** Funder runs eligibility checks	***Stage 2 -*** Proposals are reviewed by internal and/or external experts, who write report and attribute score(s)	***Stage 3 -*** Proposals are discussed in a panel, panel members vote on (final) score(s)	***Stage 4 -*** Evaluations are aggregated into a ranking or prioritisation list	***Stage 5 -*** Rankings and/or prioritisations are translated into decisions and communicated to the applicants
Call	**Eligibility checks**	**Evaluation**	**Panel discussion**	**Aggregation**	**Decision**

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- Process depends on organisation, call and funding scheme (career vs. project)
- Often considered a black-box

How do funding organisation take decisions, in practice?

The case of the SNSF, prior to 2022

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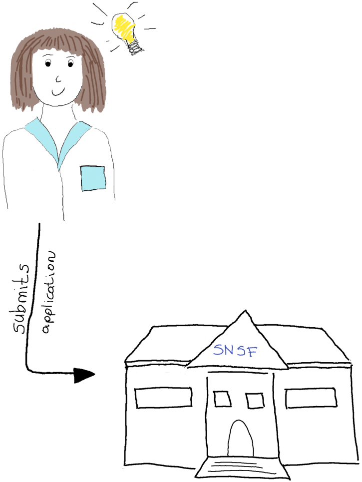

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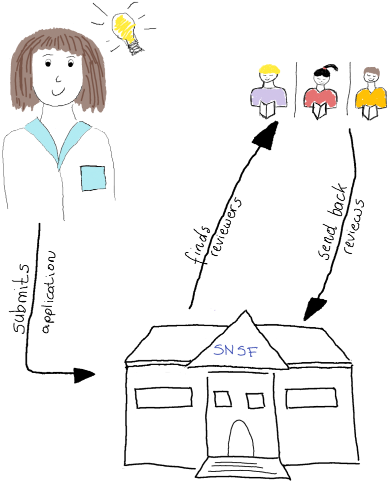

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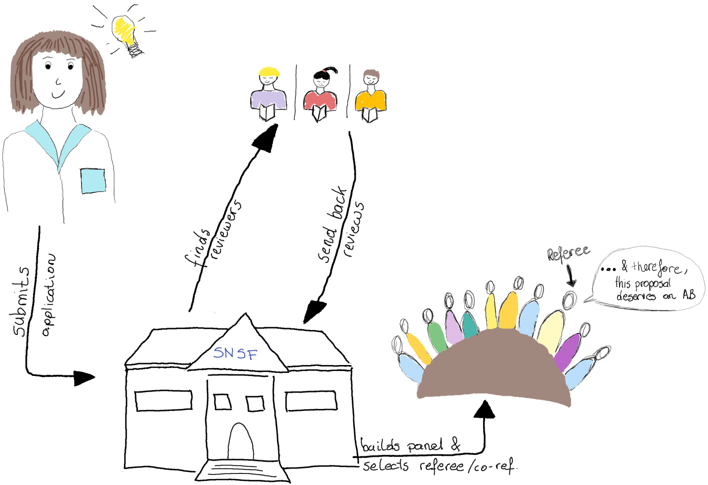

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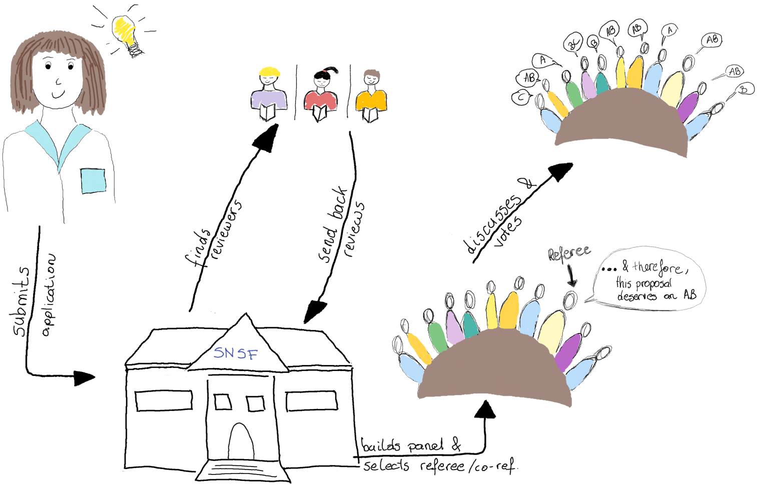


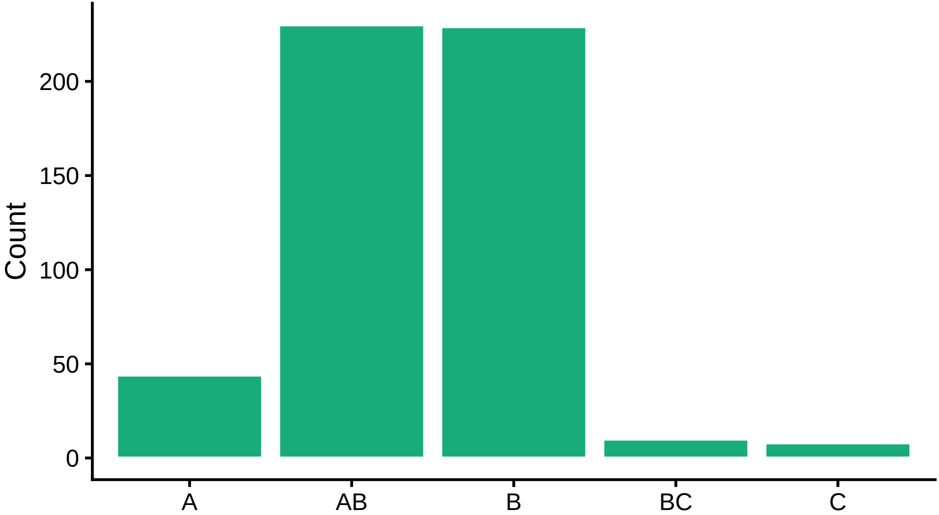
The case of the SNSF, prior to 2022

ID	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	Av
#1	C	AB	A	BC	B	AB	AB	A	AB	AB	B	4.55
#2	C	AB	A	BC	**COI**	AB	AB	A	AB	AB	B	4.6
#3	A	A	..	..	..	..	..	..	..	C	A	4.73
#4	A	AB	..	..	..	..	..	..	..	**COI**	A	5.63
#5	C	C	..	..	..	..	..	..	..	C	BC	2.33

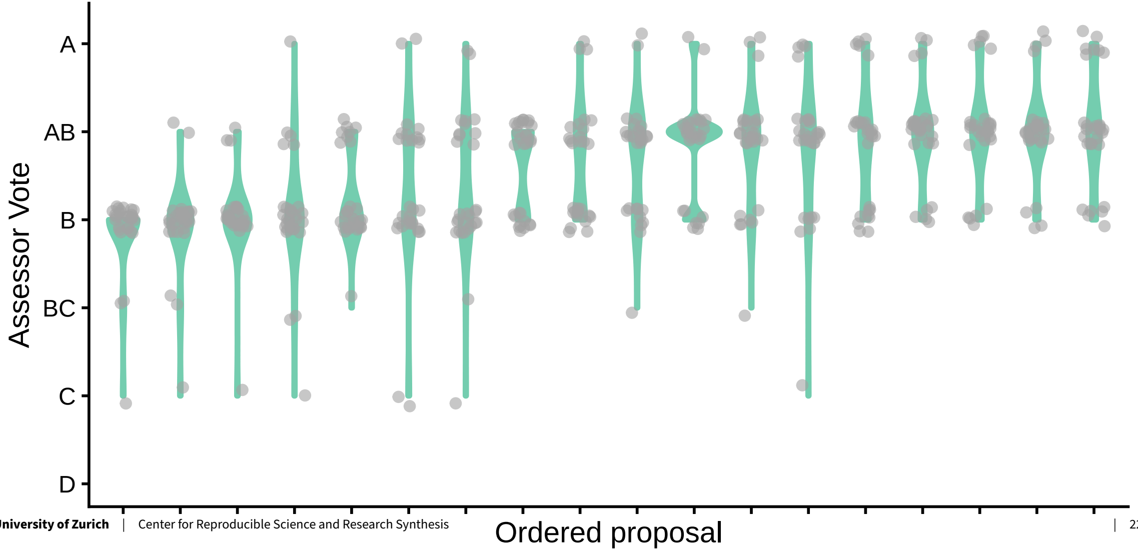
Concrete example: SNSF STEM panel & fellowships

Total nb of votes	521
Nb proposals discussed	18
Nb panel members	29
Nb fundable proposals	6
Nb COIs	1
Scale used	A, AB, B, BC, C, D

Concrete example: SNSF STEM panel & fellowships



Variation within proposals



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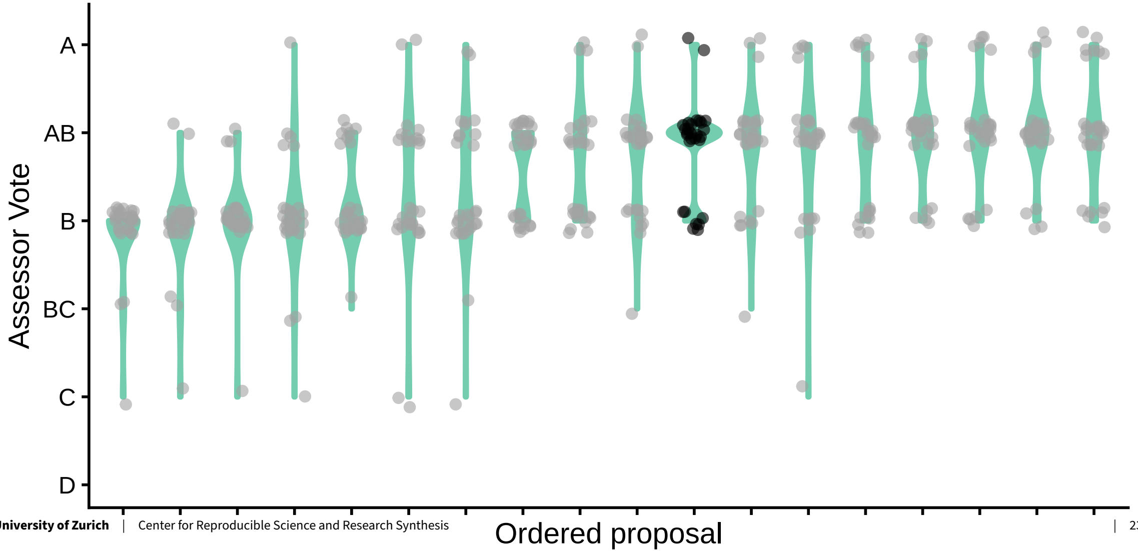

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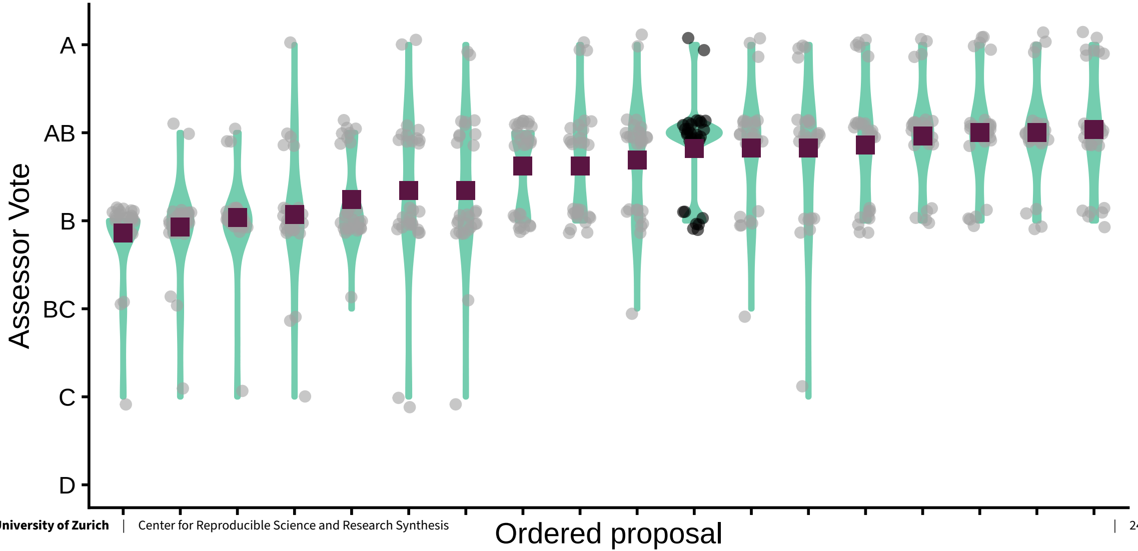
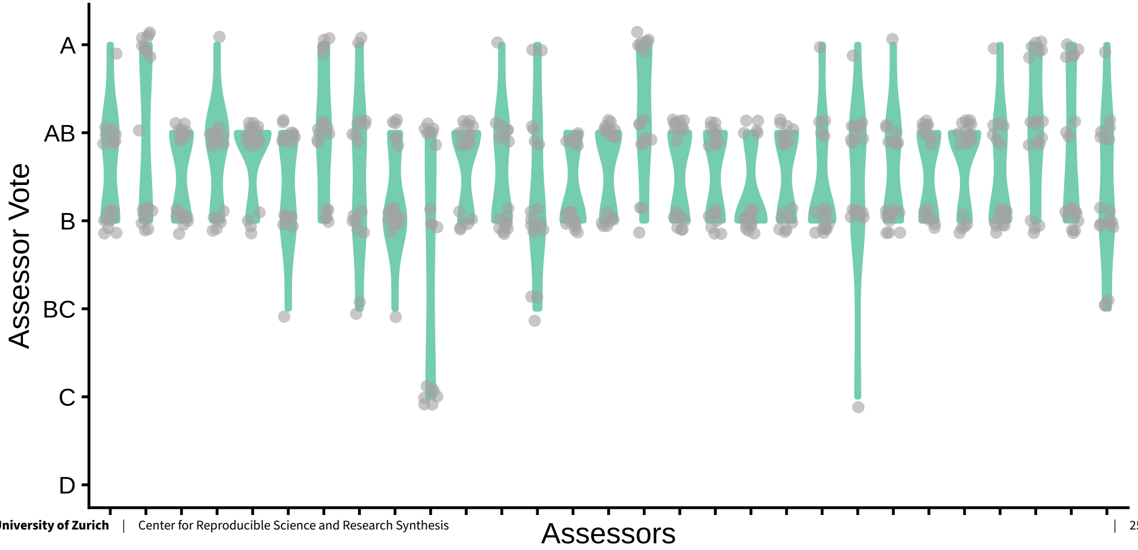


Variation within assessors



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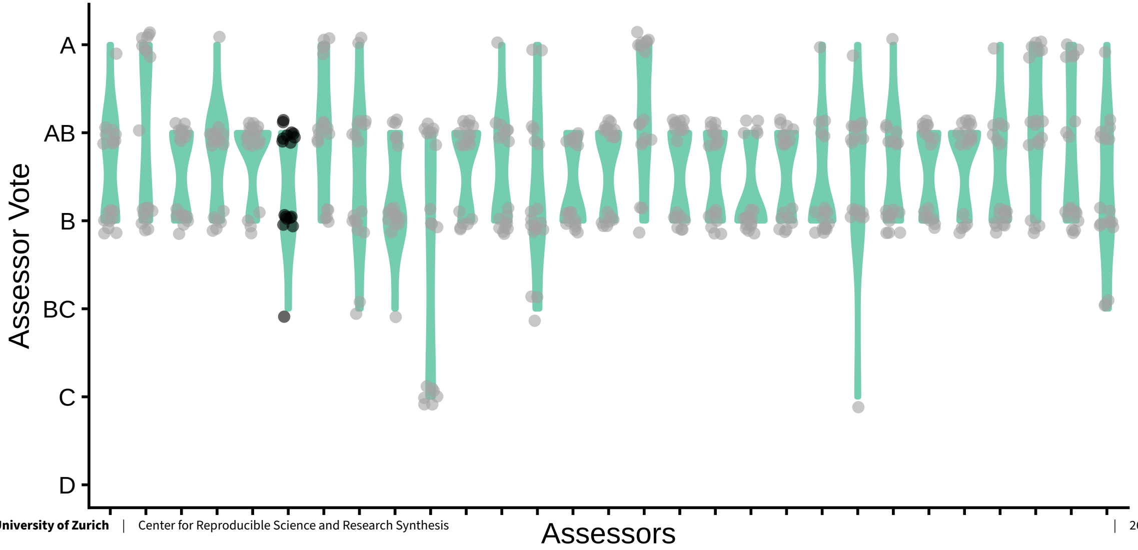

Problems for the applicant

- Decisions do not seem to be based on quality of research project
  - arbitrary
  - black box

Problems of the funder

— How to discriminate between similarly good proposals?

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- How to discriminate between similarly good proposals?
- How to account for dependency, uncertainty, variability in the data?
- But, **simplicity of averages!**

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- Averages do not work well with outliers (easy to game…)
- Disregarding variability (e.g., induced by COIs) is problematic
- Averages of ordinal data far from optimal
- Reliance on sub-optimal estimations of quality:

  $\Rightarrow$ Panel member grades are known to be unreliable

  $\Rightarrow$ Panel member grades can be biased

OK… but what now?

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- Model process “correctly”

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- Model process "correctly"
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$\Rightarrow$ Split scientific evaluation and funding decision

$\Rightarrow$ Improve transparency and reproducibility of decision-making process

Proposed solution at the SNSF

Model better

- Let’s assume a proposal of a certain quality θ_i is submitted for funding
- … [evaluation process] …

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- All panel members vote: given their expertise, biases and past experiences, voter j **estimates the quality of proposal** i

  $\rightarrow y_{ij}, i \in \{1, \ldots, n\}$ and $j \in \{1, \ldots, m\}$

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  $\rightarrow y_{ij}$, $i \in \{1, \dots, n\}$ and $j \in \{1, \dots, m\}$
- Bayesian Hierarchical Model (given some priors) for the panel votes:

$$
\begin{aligned}
y_{ij} \mid \theta_i, \lambda_{ij} &\sim N(\bar{y} + \theta_i + \lambda_{ij}, \sigma^2) \\
\theta_i &\sim N(0, \tau_\theta^2) \\
\lambda_{ij} &\sim N(\nu_j, \tau_\lambda^2).
\end{aligned}
$$

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Quantify/model uncertainty

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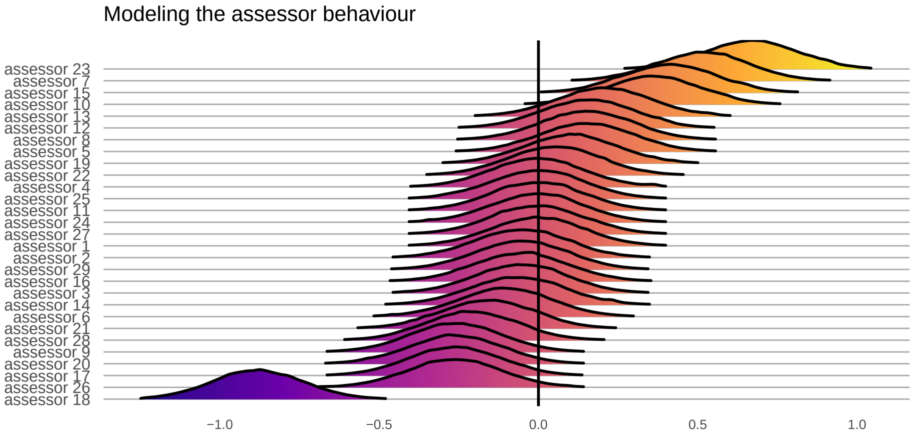

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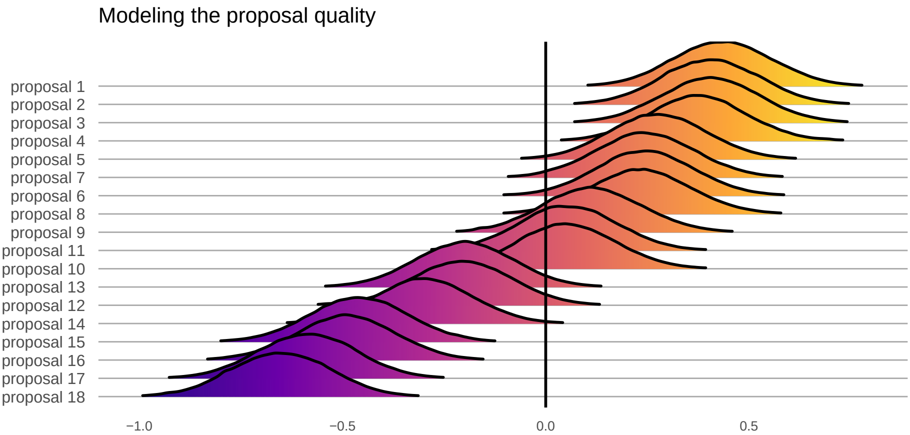

Proposed solution at the SNSF

Acknowledge uncertainty via lottery

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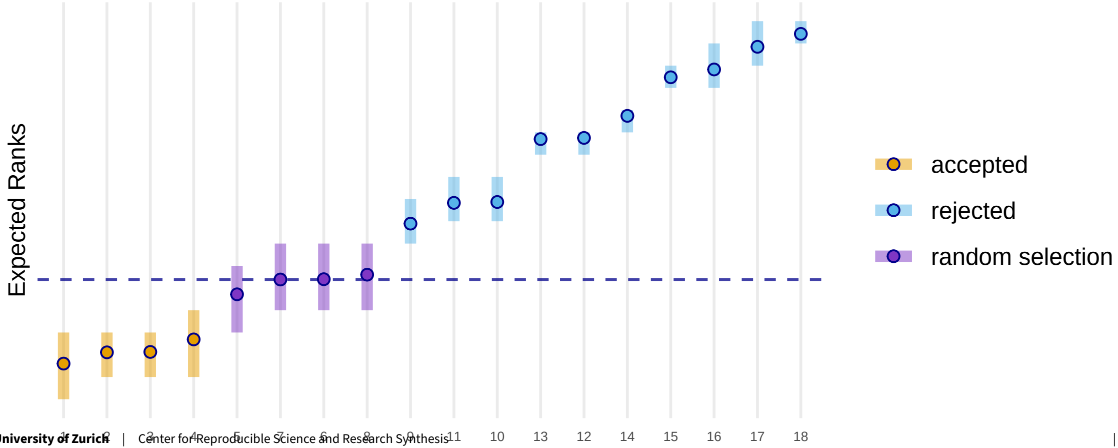

Bayesian ranking in practice

ERforResearch 4.0.0 Reference Articles Changelog

ERforResearch

This package implements the functionality to use the expected ranks (ER) in the context of research evaluation as presented in this publication. For a demonstration of the major functions check out the vignette (Demo ERforResearch).

Installation

The package can be downloaded from github using the `devtools`-package:

```
# First, you might have to install devtools from CRAN
install.packages("devtools")

# Then install ERforResearch
devtools::install_github("snsf-data/ERforResearch")
```

Requirement

The Bayesian Hierarchical Models for the ER are fitted in JAGS using `rjags`. Therefore you have to install the `rjags` from CRAN and JAGS following the instructions from here.

License

MIT + file LICENSE

Citation

Citing ERforResearch

Developers

Rachel Heyard

Author, maintainer

https://snsf-data.github.io/ERforResearch/

Bayesian ranking paper

STATISTICS AND PUBLIC POLICY
2022, VOL. 9, NO. 1, 110–121
https://doi.org/10.1080/2330443X.2022.2086190

Taylor & Francis
Taylor & Francis Group

Rethinking the Funding Line at the Swiss National Science Foundation: Bayesian Ranking and Lottery

Rachel Heyard[a,b], Manuela Ott[a], Georgia Salanti[c], and Matthias Egger[c,d]

[a]Data Team, Swiss National Science Foundation, Bern, Switzerland; [b]Center for Reproducible Science, University of Zurich, Zurich, Switzerland; [c]Institute of Social and Preventive Medicine, University of Bern, Bern, Switzerland; [d]Population Health Sciences, Bristol Medical School, University of Bristol, Bristol, UK

ABSTRACT

Funding agencies rely on peer review and expert panels to select the research deserving funding. Peer review has limitations, including bias against risky proposals or interdisciplinary research. The inter-rater reliability between reviewers and panels is low, particularly for proposals near the funding line. Funding agencies are also increasingly acknowledging the role of chance. The Swiss National Science Foundation (SNSF) introduced a lottery for proposals in the middle group of good but not excellent proposals. In this article, we introduce a Bayesian hierarchical model for the evaluation process. To rank the proposals, we estimate their expected ranks (ER), which incorporates both the magnitude and uncertainty of the estimated differences between proposals. A provisional funding line is defined based on ER and budget. The ER and its credible interval are used to identify proposals with similar quality and credible intervals that overlap with the provisional funding line. These proposals are entered into a lottery. We illustrate the approach for two SNSF grant schemes in career and project funding. We argue that the method could reduce bias in the evaluation process. R code, data and other materials for this article are available online.

ARTICLE HISTORY
Received February 2021
Accepted May 2022

KEYWORDS
Bayesian ranking; Expected rank; Funding line; Grant peer review; Modified lottery; Posterior mean

https://doi.org/10.1080/2330443X.2022.2086190

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— First implementation for SNSF.

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- With some expertise in Bayesian hierarchical models, generalisable to other funders/situations.

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- First implementation for SNSF.
- With some expertise in Bayesian hierarchical models, generalisable to other funders/situations.
- **But**, impossible to develop a one-size fits all tool.

The case of the Marie Skłodowska-Curie actions (MSCA)

PLOS ONE

RESEARCH ARTICLE

Assessing the potential of a Bayesian ranking as an alternative to consensus meetings for decision making in research funding: A case study of Marie Skłodowska-Curie actions

Rachel Heyard[1]*, **David G. Pina**[2], **Ivan Buljan**[3], **Ana Marušić**[4]

1 Center for Reproducible Science, Epidemiology, Biostatistic and Prevention Institute, University of Zurich, Zurich, Switzerland, **2** European Research Executive Agency, European Commission, Brussels, Belgium, **3** Department of Psychology, Faculty of Humanities and Social Sciences, University of Split, Split, Croatia, **4** Center for Evidence-based Medicine, Department of Research in Biomedicine and Health, University of Split School of Medicine, Split, Croatia

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Check for updates

https://doi.org/10.1371/journal.pone.0317772

Decision-making MSCA

- Individual evaluation reports (three criteria and overall score)
  - Three experts per proposal
  - Scores on scale from 1 to 100

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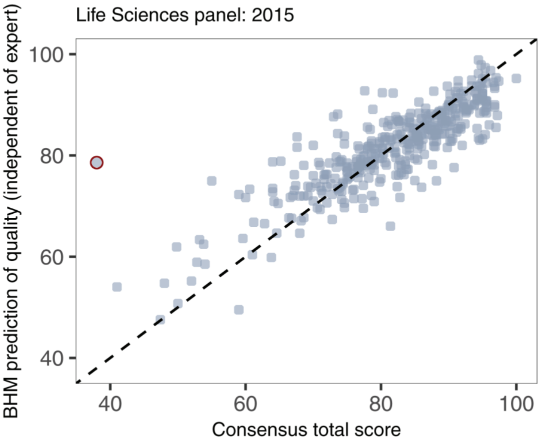
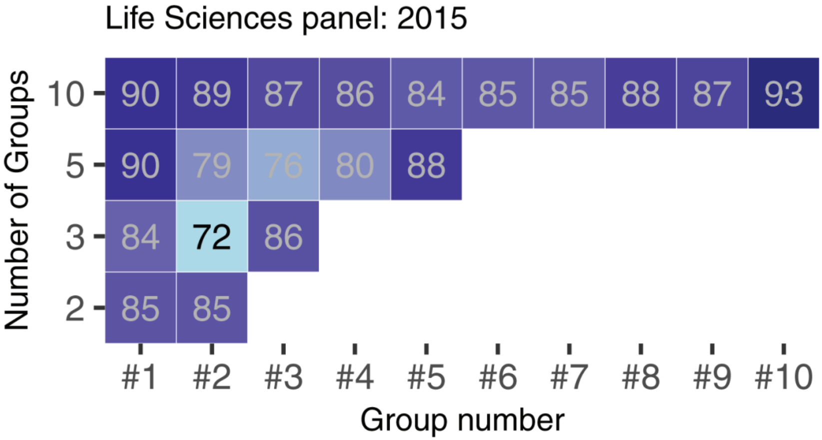
- Consensus meetings
  - Consensus report
- Ranking based on consensus report
  - Main list (accepted), Reserve list, Rejected list

How useful are individual expert scores in predicting consensus score?



Bayesian hierarchical model prediction of quality using: $\bar{y} + \theta_i$.

How useful are individual expert scores in predicting consensus ranking?



Pros and cons of Bayesian Ranking

+ Quantify uncertainty with respect to the true rank
+ Truly comparative ranking
+ Adjust for grading habits of panel members (and possible of panels)

- Higher complexity
- Longer and intense computation needed

Bayesian Ranking adopted by the SNSF and fully integrated in process in fall 2022

Lessons learned

- Bayesian Ranking is a (still imperfect) decision making tool
- Limitations and assumptions need to be clearly communicated
- Development and implementation process needs to be communicated transparently and all panel members should be included in discussion ($e.g.$ no black box)

What’s next for this ranking methodologies?

- So much we do not understand yet.

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- Using available evidence better! criteria scores, external experts vs. panel members, …

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— Using available evidence better! criteria scores, external experts vs. panel members, …

$\Rightarrow$ Taking a more holistic approach – from call design to funding decision.

Motivation for funding lotteries

Research Evaluation, 32(1), 2023, 86–100
https://doi.org/10.1093/reseval/rvac022
Advance Access Publication Date: 29 October 2022
Article

OXFORD

Peer review in funding-by-lottery: A systematic overview and expansion

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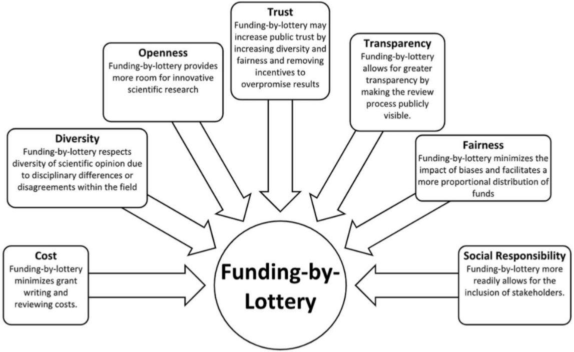

Figure 3. Motivations for funding-by-lottery.

https://doi.org/10.1093/reseval/rvac022

Effect of funding by lottery

nature communications

Article

https://doi.org/10.1038/s41467-025-65660-9

Lottery before peer review is associated with increased female representation and reduced estimated economic cost in a German funding line

Received: 6 March 2025

Accepted: 21 October 2025

Finn Luebber[1,2], Sören Krach[1,2], Frieder M. Paulus[1,2], Lena Rademacher[1,2,3] & Rima-Maria Rahal[4,5]

https://doi.org/10.1038/s41467-025-65660-9

Thank You!

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Simona Isler, Laura Lots, Julius Mattern, Anne Jorstad

Collaborators – Bayesian Ranking:
Manuela Ott, Georgia Salanti, Matthias Egger. David Pina, Ivan Buljan, Ana Marusic

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https://www.crs.uzh.ch

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Available from github: rachelhey.github.io/talks/SORTEE_webinar.

Please cite as: R Heyard "Lotteries and innovations in research funding allocation", SORTEE webinar, 2026 osf.io/ujdah.